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Multi-Domain Feature Engineering and Stacked Ensemble Learning for Epileptic Seizure Detection

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Abstract:	Automated detection of epileptic seizures from scalp electroencephalography (EEG) remains challenging due to strong inter-patient variability and the trade-off between the raw performance of deep learning models and the interpretability required for clinical adoption. Most existing EEG feature representations characterise seizure activity through aggregated statistical, spectral, or wavelet-domain descriptors, which provide only indirect information about the underlying neuronal firing dynamics. This paper proposes an interpretable, multi-domain feature engineering framework for seizure detection on the CHB-MIT Scalp EEG Database. In addition to conventional time-domain, frequency-domain, wavelet, and nonlinear descriptors, each EEG segment is used as the input current to a Leaky Integrate-and-Fire (LIF) neuron model, from which eight biologically motivated spike-train statistics -- including firing rate, burst structure, inter-spike-interval regularity, and spike-count entropy -- are extracted as an additional feature family. Two gradient-boosted classifiers, LightGBM and XGBoost, are independently tuned via Optuna and combined through an out-of-fold stacking architecture with a logistic-regression meta-learner, with the decision threshold calibrated on a held-out validation set. On a stratified test partition, the stacked ensemble achieves 98.81% accuracy, an F1-score of 0.9855, and a ROC-AUC of 0.9980, with near-symmetric precision and recall across both classes and high seizure recall in particular. Under a stricter Leave-One-Subject-Out evaluation, the framework retains 89.62% accuracy, indicating meaningful but incomplete cross-patient generalisation. These results are competitive with recently published deep-learning approaches on CHB-MIT, while retaining the interpretability, data efficiency, and computational lightness of a classical feature-driven pipeline.
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Abstract

Automated detection of epileptic seizures from scalp electroencephalography (EEG) remains challenging due to strong inter-patient variability and the trade-off between the raw performance of deep learning models and the interpretability required for clinical adoption. Most existing EEG feature representations characterise seizure activity through aggregated statistical, spectral, or wavelet-domain descriptors, which provide only indirect information about the underlying neuronal firing dynamics. This paper proposes an interpretable, multi-domain feature engineering framework for seizure detection on the CHB-MIT Scalp EEG Database. In addition to conventional time-domain, frequency-domain, wavelet, and nonlinear descriptors, each EEG segment is used as the input current to a Leaky Integrate-and-Fire (LIF) neuron model, from which eight biologically motivated spike-train statistics – including firing rate, burst structure, inter-spike-interval regularity, and spike-count entropy – are extracted as an additional feature family. Two gradient-boosted classifiers, LightGBM and XGBoost, are independently tuned via Optuna and combined through an out-of-fold stacking architecture with a logistic-regression meta-learner, with the decision threshold calibrated on a held-out validation set. On a stratified test partition, the stacked ensemble achieves 98.81% accuracy, an F1-score of 0.9855, and a ROC-AUC of 0.9980, with near-symmetric precision and recall across both classes and high seizure recall in particular. Under a stricter Leave-One-Subject-Out evaluation, the framework retains 89.62% accuracy, indicating meaningful but incomplete cross-patient generalisation. These results are competitive with recently published deep-learning approaches on CHB-MIT, while retaining the interpretability, data efficiency, and computational lightness of a classical feature-driven pipeline.

Keywords: EEG-based seizure detection; multi-domain features; spiking neural network; ensemble learning; explainable AI; SHAP.

1 Introduction

Epilepsy is a chronic neurological disorder characterised by recurrent, unprovoked seizures arising from abnormal and excessive neuronal discharges in the brain [20, 17, 16]. It affects more than 50 million individuals worldwide, with an annual incidence ranging from 23 to 100 cases per 100,000 people, imposing substantial clinical, psychological, and socioeconomic burden [34, 47]. Despite considerable advances in antiepileptic medication, dietary therapy, neurostimulation, and resective surgery, approximately one-third of patients with epilepsy remain drug-resistant, experiencing persistent seizures that significantly reduce quality of life [48, 30]. The development

of reliable automated seizure detection and prediction systems is therefore an important objective for improving diagnosis, monitoring, and clinical decision-making. Electroencephalography (EEG) remains the gold-standard non-invasive modality for recording cortical electrical activity and is extensively used in epilepsy diagnosis and long-term monitoring [27, 2]. EEG signals exhibit highly nonlinear, non-stationary, multiscale dynamics arising from complex neuronal interactions, and are commonly classified into four physiological states: interictal, preictal, ictal, and postictal [21, 29]. Manual inspection of prolonged EEG recordings by trained clinicians is laborious, time-consuming, and susceptible to inter-observer variability, motivating the development of automated computational frameworks capable of operating reliably in real-time clinical settings.

Early automated seizure-detection systems relied on handcrafted statistical, spectral, wavelet, and nonlinear features combined with conventional classifiers. [33] applied nonlinear signal analysis to characterize EEG across different mental states; [35] combined discrete wavelet transform features with a support vector machine for EEG classification; and [38] proposed a time-frequency-localised orthogonal wavelet filter bank for automatic detection of focal EEG signals. [46] further considered the practical constraint of power-efficient patient monitoring alongside classification accuracy. These approaches established that carefully engineered features can separate ictal from inter-ictal activity effectively, but their performance depended heavily on feature quality and generalised poorly across patients and recording conditions. The emergence of deep learning shifted the field toward end-to-end feature learning directly from raw EEG. [5] proposed a deep convolutional neural network for automated seizure detection and diagnosis; [3] combined convolutional and bidirectional LSTM layers to jointly capture spatial and temporal seizure dynamics; [45] applied convolutional architectures to both intracranial and scalp EEG for seizure prediction; and [49] developed a fully end-to-end CNN-based detector. While these models achieve strong raw performance, [14] and the review of [37] highlight that deep architectures typically require large labelled datasets, incur substantial computational cost, and exhibit black-box behaviour that limits clinical trust and adoption.

More recently, self-attention mechanisms have been applied to EEG seizure analysis to model long-range temporal dependencies and cross-channel interactions. [15] proposed a transformer-based seizure predictor; [36] developed a transformer model for seizure prediction from scalp EEG; and [19] introduced Seizure-former, an attention-gated transformer architecture for seizure prediction. These works report state-of-the-art discrimination but inherit the same data-efficiency and interpretability limitations as other deep architectures. Parallel to these developments, spiking neural networks (SNNs), often described as the third generation of neural network models, process information through discrete spike events rather than continuous activations, naturally encoding temporal dynamics while offering substantially lower computational and energy requirements than conventional artificial neural networks [32]. Despite this promise, biologically grounded spike-based feature representations remain comparatively underexplored specifically for EEG seizure detection, where most existing work still operates on continuous-valued statistical, spectral, or wavelet descriptors rather than on an explicit spike-domain transformation of the signal.

Motivated by the interpretability and data-efficiency limitations of deep architectures, a number of recent studies have revisited carefully engineered feature representations combined with ensemble learning and explainable AI (XAI). This trend is documented across several EEG-based diagnostic domains. In EEG-based schizophrenia classification, a distinct but methodologically related diagnostic task, [7] combined multi-scale and adaptive feature selection with deep learning; [6] used image-encoded EEG representations with wrapper-based deep feature selection; [24] proposed a hybrid graph-convolutional GCN–LSTM model; [26] developed a brain-lobe-specific explainable framework; [42] fused EEG microstate and empirical-mode-decomposition features;

and [43] combined variational mode decomposition with Bayesian-optimized machine learning. Collectively, these studies indicate that, across EEG-based neurological and psychiatric diagnosis more broadly, well-engineered feature representations combined with ensemble learning and XAI can match deep-learning performance while offering substantially greater transparency, a pattern this paper investigates specifically for epileptic seizure detection.

Within seizure detection proper, several recent CHB-MIT-based studies provide direct benchmarks for the present work. [50] proposed a hybrid convolutional sparse autoencoder–GRU (CSAE–GRU) framework with two-stage class balancing; [28] applied a graph convolutional network to frequency-band-decomposed EEG; [12] developed a single/four-channel patient-specific detector with clinical confirmation of seizure locations; [11] combined a 1D CNN–LSTM architecture with discrete wavelet transform features; a vision-transformer approach [4] recast EEG as spectrogram images for Mix-ViT/Swin-based classification; [8] proposed an ensemble deep-transfer-learning detector; and [18] introduced EpiNet, a hybrid machine-learning model evaluated on a 500-patient dataset. These studies are used as direct comparators for the proposed framework in Section 3.

Most existing EEG features, including statistical moments, spectral band powers, entropy measures, nonlinear complexity metrics, and wavelet energies, characterize EEG activity through aggregated amplitude, frequency, or scale-domain statistics. While informative, such descriptors provide only indirect information about the neuronal firing processes responsible for seizure generation. Epileptic seizures are fundamentally characterized by abnormal hypersynchronous neuronal firing, producing the rhythmic bursts and oscillatory activity directly observable in scalp EEG [22, 44]. Motivated by this observation, we hypothesize that transforming EEG segments into biologically meaningful spike trains can reveal complementary information unavailable in conventional signal representations. To this end, each EEG segment is used as the input current to a Leaky Integrate-and-Fire (LIF) neuron model [9], from which a collection of spike-domain descriptors, firing rate, burst structure, inter-spike interval statistics, and spike-count entropy is extracted. These descriptors explicitly characterize neuronal firing dynamics and provide information that is largely orthogonal to conventional statistical, spectral, nonlinear, and wavelet features.

Building on this concept, this paper proposes an interpretable seizure detection framework that integrates multi-domain EEG feature extraction, biologically inspired spike-domain modelling and stacked ensemble learning. The main contributions of this work are as follows.

1. A comprehensive multi-domain EEG representation combining statistical time-domain, frequency-domain, wavelet-domain, nonlinear complexity, and biologically inspired spike-domain descriptors, yielding 42 features items per channel (Section 2.4).

2. A stacked ensemble learning architecture combining LightGBM and XGBoost, independently tuned via Optuna, and combined through an out-of-fold logistic-regression meta-learner for robust seizure classification (Section 2.5).

3. A SHAP-based explainability framework that provides feature-level interpretation of model predictions and links important EEG biomarkers to established neurophysiological mechanisms associated with epileptic seizures.

4. A novel LIF neuron-based feature engineering framework that transforms EEG segments into spike trains and extracts firing-rate, burst, inter-spike-interval, and entropy descriptors.

5. Comprehensive evaluation under both subject-mixed and subject-independent (LOSO) protocols, providing insight into the generalisation capability of the proposed seizure-detection framework across previously unseen patients.

The remainder of this paper is organised as follows. Section 2 presents the proposed methodology, including EEG preprocessing, multi-domain feature extraction, spike-domain modelling, and ensemble learning. Section 3 reports the numerical results and comparative analysis against recent CHB-MIT-based studies and some discussion is given in Section 4. Finally, Section 5 concludes the paper and outlines future research directions.

2 Materials and Methods

This section describes the dataset and preprocessing pipeline in Section 2.1. The multi-domain feature extraction framework spanning time, frequency, wavelet, nonlinear, and spike-domain descriptors present in Section 2.2 with the biologically motivated LIF spike-train feature construction in Sections 2.3, the resulting unified feature representation in Section 2.4, the stacked ensemble classification and SHAP-based interpretability framework used for seizure detection in Section 2.5.

2.1 Dataset Description

All experiments utilise the **CHB-MIT Scalp EEG Database** [40], a widely adopted benchmark for seizure detection and prediction research, recorded at Children's Hospital Boston and distributed through PhysioNet [23]. The corpus comprises continuous scalp EEG recordings from 22 paediatric subjects (ages 1.5–22 years) with intractable epilepsy, sampled at 256 Hz using the International 10–20 electrode system, spanning 664 European Data Format (EDF) files with 916 annotated seizure events. Precise seizure onset and offset timestamps are provided in per-subject summary files accompanying the recordings. Figure 1 shows an example of EEG segment with seizure annotation.

Raw recordings are pre-processed through three operations applied in sequence. First, files whose native sampling frequency deviates from 256 Hz are resampled via polyphase filtering to enforce a uniform temporal resolution across subjects. Second, a zero-phase finite impulse response (FIR) bandpass filter with pass-band $[0.5, 70]$ Hz removes slow baseline drift and high-frequency artefacts while retaining clinically relevant EEG rhythms. Third, a 60 Hz notch filter suppresses power-line interference intrinsic to North American recording environments. Recordings with fewer than the retained channel count are zero-padded to maintain a uniform tensor shape $\mathbf{S} \in \mathbb{R}^{C \times T}$ across subjects.

Preprocessed recordings are segmented into fixed-length, non-overlapping analysis windows of 2 s duration ($L_w = 512$ samples at 256 Hz). Seizure annotations are extended by 10 s before onset and 5 s after offset to include the preictal and postictal transitions, capturing clinically relevant dynamics surrounding each ictal event. A two-pass segmentation strategy is used to avoid sampling bias between the highly imbalanced ictal and inter-ictal classes: ictal-associated segments are extracted with 90% overlap (stride = 51 samples) to maximise the number of examples drawn from the comparatively rare seizure periods, while inter-ictal segments are extracted independently with 50% overlap (stride = 256 samples). Because the two classes are segmented in separate passes rather than from a single sliding window, every valid temporal position is covered and no inter-ictal region is inadvertently skipped.

To remove class-imbalance as a confound in downstream model training and evaluation, the majority class (inter-ictal) is randomly undersampled to match the number of available ictal-associated segments. No synthetic samples are generated at any stage; all examples correspond to real, recorded EEG. This segmentation and balancing procedure yields the final class-balanced segment set used throughout this study.

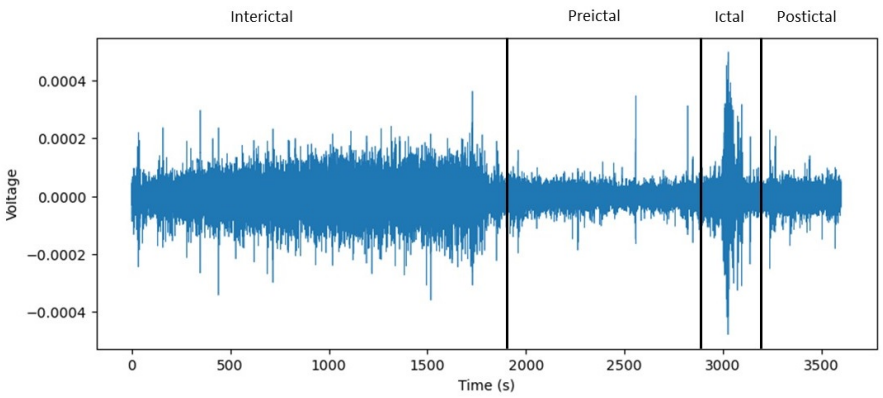


Figure 1: Example segment of a preprocessed EEG recording from the CHB-MIT database.

2.2 Multi-Domain Feature Extraction

To capture the characteristics of epileptic EEG activity from complementary perspectives, four feature families are extracted from each preprocessed channel segment: time-domain statistics, frequency-domain spectral descriptors, wavelet-domain multi-resolution energies, and nonlinear complexity measures. Each family targets a distinct aspect of the signal, its amplitude distribution and waveform structure, its spectral energy distribution, its localised time-frequency behaviour, and its chaotic/nonlinear dynamics, respectively, so that the resulting representation captures discriminative seizure-related patterns that no single domain can fully express on its own.

2.2.1 Time-Domain Features

Time-domain analysis provides direct insight into the amplitude distribution, temporal variability, and morphological characteristics of EEG signals. Epileptic seizures are typically associated with abrupt amplitude changes, sharp transients, and irregular oscillatory patterns, which can be effectively captured through statistical and dynamical descriptors in the time domain.

In this study, a set of discriminative time-domain features is extracted to characterize these properties. Specifically, higher-order statistical measures such as skewness and kurtosis are used to quantify the asymmetry and peakedness of the signal distribution. The zero-crossing rate (ZCR) is included as a proxy for frequency-related behavior, reflecting the rate of sign changes in the signal. Measures of signal variability and structure, such as peak-to-peak amplitude, interquartile range (IQR), and curve length (the sum of absolute differences between successive samples), are employed to capture amplitude fluctuations and signal roughness. In addition, Hjorth parameters are utilized to describe the dynamic behavior of EEG signals. Hjorth mobility reflects the mean frequency content, while Hjorth complexity measures deviations from a pure sinusoidal waveform, providing an estimate of signal irregularity. These features jointly capture complementary aspects of EEG dynamics, including energy distribution, temporal variation, and structural complexity.

The mathematical definitions of these eight features are summarized in Table 1, where $x(t)$ denotes the EEG signal and N represents the number of samples.

Table 1: List of extracted features grouped by time-domain.

No.	Feature	Description
1	Skewness	$\frac{\sum(x-\bar{x})^3}{\sigma^3}$: 3rd standardised moment-varies per channel
2	Kurtosis	$\frac{\sum(x-\bar{x})^4}{\sigma^4}$: 4th standardised moment-spike sensitivity
3	Zero-crossing rate	$\frac{1}{N}\sum \mathbb{I}(x_n x_{n+1} < 0)$ - Sign changes
4	Peak-to-Peak	peak-to-peak amplitude range
5	Curve Length	$\Sigma\lvert\Delta x\rvert$ - total signal trajectory
6	Hjorth Mobility	$\sqrt{\mathrm{Var}(x')/\mathrm{Var}(x)}$ - mean frequency proxy
7	Hjorth Complexity	$\frac{\mathrm{Mob}(x')}{\mathrm{Mob}(x)}$ – Irregularity — bandwidth complexity
8	Interquartile range	Q3–Q1 - robust amplitude spread

2.2.2 Frequency-Domain Features

Frequency-domain analysis provides critical insight into the spectral composition of EEG signals, enabling the identification of abnormal rhythmic activity associated with epileptic seizures. Unlike time-domain representations, spectral analysis reveals how signal energy is distributed across different frequency bands, which is particularly important for characterizing pathological brain dynamics.

Frequency-domain features are derived using Welch's Power Spectral Density (PSD) estimation method with a 128-point Hann window, which improves spectral estimation stability by averaging periodograms over overlapping segments, thereby reducing variance and noise sensitivity. Let $P(f)$ denote the estimated PSD of the EEG signal, representing the distribution of signal power over frequency. Band power features are computed over standard EEG frequency ranges, including delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz), and gamma (30–70 Hz), corresponding to well-established neurophysiological rhythms whose energy redistribution often indicates abnormal brain activity. The mean frequency is computed as the center of mass of the PSD, providing a global measure of frequency content, while the dominant frequency, defined as the frequency at which the PSD attains its maximum value, reflects the most prominent oscillatory component in the signal.

Spectral entropy quantifies the complexity and disorder of the frequency distribution: lower values indicate concentrated spectral energy, while higher values correspond to more uniform, disordered distributions often observed during seizure activity. The alpha-to-beta power ratio is included as a physiologically meaningful indicator of cortical activation and arousal state. Spectral flatness distinguishes tonal from noise-like signals by comparing the geometric and arithmetic means of the PSD, and the spectral edge frequency (95%) gives the frequency below which 95% of total spectral power is contained. The mathematical definitions of these eleven features are provided in Table 2.

2.2.3 Wavelet-Domain Features

Time-frequency analysis is essential for EEG signal processing due to the inherently non-stationary nature of brain activity. The Discrete Wavelet Transform (DWT) captures localized signal characteristics across multiple temporal and spectral scales, providing simultaneous time and frequency resolution that classical Fourier-based methods lack, particularly useful for detecting transient and abrupt changes associated with epileptic seizures.

The Daubechies-4 (db4) wavelet is used for its compact support and strong ability to represent sharp transitions and oscillatory patterns in EEG signals. A multi-level decomposition up

Table 2: List of extracted features grouped by frequency-domain.

No.	Feature	Description
9	Delta power	(0.5-4) Hz band power - Low-frequency activity
10	Theta power	(4–8) Hz band power - Low-frequency activity
11	Alpha power	(8–13) Hz band power - Rhythm stability
12	Beta power	(13–30) Hz band power - Cognitive activity
13	Gamma power	(30–70) Hz band power - Seizure activity
14	Spectral Centroid	$\frac{\sum fP(f)}{\sum P(f)}$ – Spectral center, PSD centre of mass
15	Dominant frequency	$\arg\max \lvert FFT(s)\rvert$ – Main rhythm, frequency of PSD peak
16	Spectral Entropy	$-\sum P(f)\log P(f)$ - Spectral disorder, normalised PSD
17	Alpha vs Beta	P_α/P_β — cortical arousal index
18	Spectral Flatness	$\frac{\exp(\frac{1}{N}\sum \ln P)}{\frac{1}{N}\sum P}$ – Noise vs tone, geometric/arithmetic PSD ratio
19	Spectral edge	95th-percentile frequency

to level 4 yields one approximation coefficient set and multiple detail coefficient sets corresponding to different frequency bands. Sub-band energies are computed from the detail coefficients, and the total wavelet energy is calculated as their sum. Wavelet entropy, computed from the normalized sub-band energies, reflects the degree of disorder across scales; the high-frequency energy ratio captures the proportion of energy in the finer-scale detail coefficients, associated with rapid oscillations during seizure activity; and the average standard deviation (std) across sub-bands captures the dispersion of wavelet coefficients. The mathematical definitions of these eight features are summarized in Table 3.

Table 3: List of extracted features grouped by wavelet.

No.	Feature	Description
20	Wavelet-1	detail d4 energy (lowest detail / coarsest)
21	Wavelet-2	detail d3 energy
22	Wavelet-3	detail d2 energy
23	Wavelet-4	detail d1 energy (highest detail / finest)
24	Wavelet Entropy	$-\sum p_k \log p_k$, Multi-scale randomness
25	Total Wavelet energy	$\sum E_j$, - total detail energy
26	High-frequency Wavelet	$(E_{d1}+E_{d2})/E_{\text{tot}}$, — high-freq proportion
27	Wavelet dispersion	mean std of detail coefficient vectors

2.2.4 Nonlinear Dynamical Features

EEG signals exhibit complex nonlinear and chaotic dynamics that cannot be fully characterized using linear time- or frequency-domain analyses. Epileptic seizures are associated with abrupt transitions, increased irregularity, and changes in signal complexity, motivating a set of nonlinear features that provide complementary information about the intrinsic dynamics of the signal.

Permutation entropy quantifies the complexity of temporal patterns by measuring the diversity of ordinal structures within the signal; it is robust to noise and computationally efficient, making it suitable for real-time EEG analysis. The Petrosian fractal dimension estimates the fractal characteristics of the signal from the rate of sign changes in its derivative. The nonlinear

energy operator (NLEO) captures instantaneous energy changes, making it particularly sensitive to transient events such as epileptic spikes; both the mean and standard deviation of the NLEO are computed. The Hilbert transform yields the analytic signal representation, from which the mean instantaneous amplitude envelope and the standard deviation of the unwrapped instantaneous phase are derived. Finally, the crest factor, the ratio of peak amplitude to signal RMS power, captures spike-like behavior and sharp transients characteristic of epileptic discharges. The mathematical definitions of these seven features are presented in Table 4.

Table 4: List of extracted features grouped by non-linear domain.

No.	Feature	Description
28	Permutation Entropy	$-\sum p(\pi)\log p(\pi)$ - Complexity, ordinal pattern entropy (order 3)
29	Petrosian	fractal dimension via sign-change count
30	Nonlinear mean	mean Nonlinear Energy Operator $y[n] = x[n]^2 - x[n-1]x[n+1]$
31	Nonlinear deviation	std of NLEO - burst irregularity
32	Hilbert amplitude	mean instantaneous amplitude envelope
33	Hilbert phase	std of unwrapped instantaneous phase
34	Crest Factor	$\frac{\max\lvert s_n\rvert}{\text{RMS}}$ - Peak-to-RMS, spike-to-power ratio

2.3 Spike-Domain Feature

The LIF model is the simplest biophysically motivated single-neuron model that retains the essential mechanism of action potential generation [9]. The sub-threshold membrane potential dynamics obey:

$$\tau_m \frac{dV(t)}{dt} = -\big(V(t) - V_{\text{rest}}\big) + R\,I(t), \tag{1}$$

where $V(t)$ is the membrane potential, τ_m is the membrane time constant, V_{rest} is the resting potential, R is the membrane resistance, and $I(t)$ is the input current. A spike is generated whenever $V(t)$ reaches the threshold V_{th}, after which the membrane is immediately reset:

$$V(t^+) = V_{\text{rest}} \qquad \text{if } V(t) \geq V_{\text{th}}.$$

In discrete time with step $\Delta t = 1/f_s$, Equation (1) integrates to the exact solution

$$V[t] = V[t-1] \cdot \underbrace{\exp\left(-\frac{\Delta t}{\tau_m}\right)}_{\text{decay}} + I[t],$$

where the input current $I[t]$ is the z-score normalised EEG sample at time step t:

$$I[t] = \frac{x[t] - \mu_x}{\sigma_x + \varepsilon},$$

with μ_x and σ_x the sample mean and standard deviation of the channel segment, and $\varepsilon = 10^{-9}$ a numerical stability constant. Z-score normalisation ensures scale invariance across subjects and electrode positions while preserving the relative temporal dynamics of the signal.

The LIF model parameters used in this work are summarised in Table 5. The membrane time constant $\tau_m = 20$ ms is chosen to match the integration time of cortical pyramidal neurons [41], the primary generators of ictal discharges. The firing threshold $V_{\text{th}} = 1.0$ (in normalised units) is selected so that the neuron fires when the input current exceeds one standard deviation – corresponding to the amplitude fluctuations characteristic of ictal EEG [39].

Table 5: Leaky Integrate-and-Fire neuron parameters.

Parameter	Symbol	Value	Physiological Basis
Membrane time constant	τ_m	20 ms	Cortical pyramidal neuron [41]
Firing threshold	V_{th}	1.0 (normalised)	1σ of input signal
Resting potential	V_{rest}	0.0	Reference baseline
Membrane resistance	R	1.0	Absorbed into normalised input
Burst ISI threshold	δ_{burst}	40 ms	Cortical burst criterion [25]
Spike-count bin width	Δt_{bin}	50 ms	Temporal resolution for entropy

Spike-Train Feature

Following LIF simulation for each channel segment, a spike-time set $\mathcal{S} = \{t_k\}_{k=1}^{n_s}$ is obtained, where n_s is the total spike count and t_k denotes the k-th spike time in samples. Eight spike-train statistics are derived from $\mathcal{S}$ and the membrane potential trajectory $V[t]$, as defined in Table 6. The inter-spike interval (ISI) sequence is

$$\text{ISI}_k = \frac{t_{k+1} - t_k}{f_s}, \qquad k = 1, \ldots, n_s - 1, \tag{2}$$

with mean $\bar{\text{ISI}}$, standard deviation σ_{ISI}, and coefficient of variation $\text{CV}_{\text{ISI}} = \sigma_{\text{ISI}}/(\bar{\text{ISI}} + \varepsilon)$. The CV is a dimensionless regularity index: $\text{CV} \approx 0$ indicates perfectly regular (clock-like) firing, while $\text{CV} \approx 1$ characterises Poisson-like random firing. Ictal EEG typically produces rhythmic, high-rate bursts with $\text{CV} \ll 1$, whereas inter-ictal background activity produces sparse, irregular spikes with $\text{CV} \approx 1$ [13].

A spike at time t_{k+1} is classified as part of a *burst* if its preceding ISI satisfies $\text{ISI}_k < \delta_{\text{burst}}$ (50 ms, Table 5). The burst index β_{idx} quantifies the proportion of spikes participating in bursts, and $\bar{b}$ is the mean number of spikes per burst episode, both substantially elevated during ictal activity [44]. The spike-count entropy H_{sp} is computed over a histogram of spike counts in non-overlapping 50 ms bins $\{b_j\}$:

$$H_{\text{sp}} = -\sum_j p_j \ln p_j, \quad p_j = \frac{b_j}{\sum_j b_j + \varepsilon}. \tag{3}$$

Low entropy indicates periodic, predictable spike bursts (ictal), while high entropy indicates temporally diffuse, unpredictable firing (inter-ictal), consistent with information-theoretic analyses of ictal dynamics [31]. The silent fraction ξ measures the proportion of time the membrane potential remains sub-threshold, capturing the complementary notion of quiescence between bursts. Together, these eight statistics provide a compact signature of the EEG's temporal fine structure from a neural-coding perspective, complementing the four classical feature families of Section 2.2.

2.3.1 Theoretical Motivation

The LIF spike-train feature set constitutes a novel contribution to EEG-based seizure detection for three reasons. Firstly, the eight SNN features are derived from a fundamentally different signal representation, a nonlinear, threshold-triggered spike train, rather than from the continuous amplitude waveform. Amplitude statistics (skewness, kurtosis) describe the distribution of instantaneous values; spectral features describe frequency composition; wavelet features describe

Table 6: SNN/LIF spike-train features (features 35–42). $\mathcal{S}$: spike-time set; $n_s = |\mathcal{S}|$; ISI_k: inter-spike interval sequence; $V[t]$: membrane potential trajectory; L_w: window length (samples); f_s: sampling frequency; ε: numerical stability constant.

#	Feature	Definition & Description
35	Spike Rate	$r_s = \dfrac{n_s}{L_w/f_s}$: Ictal EEG drives rapid, sustained neuronal firing; inter-ictal EEG produces sparse spikes [22].
36	Mean ISI	$\bar{\mathrm{ISI}} = \dfrac{1}{n_s-1}\sum_{k=1}^{n_s-1} \mathrm{ISI}_k$: High-frequency ictal bursts reduce the time between consecutive spikes dramatically [13].
37	ISI Std	$\sigma_{\mathrm{ISI}} = \sqrt{\dfrac{1}{n_s-1}\sum(\mathrm{ISI}_k - \bar{\mathrm{ISI}})^2}$: Synchronised ictal bursts produce regular inter-spike intervals with low variability.
38	Coefficient of Variation	$\mathrm{CV}_{\mathrm{ISI}} = \dfrac{\sigma_{\mathrm{ISI}}}{\bar{\mathrm{ISI}} + \varepsilon}$. Near zero for ictal (rhythmic), near one for inter-ictal (Poisson-like); a dimensionless regularity index sensitive to burst structure [13].
39	Burst Index	$\beta_{\mathrm{idx}} = \dfrac{\lvert\{k : \mathrm{ISI}_k < \delta_{\mathrm{burst}}\}\rvert}{n_s}$ ($\delta_{\mathrm{burst}} = 40$ ms). Ictal hypersynchrony generates densely clustered spike bursts; inter-ictal spikes are isolated [44].
40	Mean Burst Length $\bar{b}$	Mean number of consecutive spikes in burst episodes (run-length over β_{idx} assignments). Sustained ictal bursts contain more spikes per episode than sporadic inter-ictal events [25].
41	Silent Fraction	$\xi = \Pr\big(V[t] < V_{\mathrm{th}}\big) = \dfrac{1}{L_w}\sum_t \mathbf{1}[V[t] < V_{\mathrm{th}}]$. Persistent ictal input keeps the neuron near threshold, reducing silent (sub-threshold) intervals.
42	Spike-Count Entropy	$H_{\mathrm{sp}} = -\sum_j p_j \ln p_j$ over 50 ms spike-count bins ((3). Low during seizures (periodic), high inter-ictally (diffuse) [31].

multi-scale energy. None of these can characterise *when* a neuron would fire, how fast it would fire, or how regularly it would burst precisely what the SNN features encode. The orthogonality of spike-domain descriptors suggests they may provide complementary information beyond conventional EEG features.

Secondly, the LIF model is the canonical minimal model of cortical neuron dynamics [9]. Using it as a feature extractor is equivalent to asking what a real neuron embedded in the EEG signal would infer about the network state. During a seizure, the expected answer is rapid synchronised bursting with short ISI, low CV, high burst index, and low spike entropy, precisely the statistical signature of ictal hypersynchrony observed in electrophysiology.

Finally, the discrete-time LIF simulation in (2.3) requires only $L_w = 512$ floating-point operations per channel (one multiply-add per sample step), with no matrix inversion, FFT, or wavelet filter bank. The subsequent spike-train statistics are computed from the spike-time array in $\mathcal{O}(n_s)$ time, where $n_s \ll L_w$. The per-channel overhead of the SNN feature block is therefore negligible compared with the Welch PSD ($\mathcal{O}(L_w \log L_w)$) and DWT ($\mathcal{O}(L_w)$) already present in the pipeline.

Table 7: Expected SNN feature values for ictal vs. inter-ictal EEG, and direction of change. ↑: higher in ictal; ↓: lower in ictal.

Feature	Ictal	Inter-ictal	Direction	Key Reference
Spike rate r_s	High ($\gg$1 Hz)	Low ($<$1 Hz)	↑	[22]
Mean ISI	Short (10–50 ms)	Long (100–500 ms)	↓	[13]
ISI Std	Low	High	↓	[13]
ISI CV	≈ 0 (regular)	≈ 1 (Poisson)	↓	[13]
Burst index	High ($>$0.5)	Low ($<$0.1)	↑	[44]
Mean burst length	Long ($>$3 spikes)	Short (1–2 spikes)	↑	[25]
Spike entropy	Low (periodic)	High (random)	↓	[31]

2.4 Unified Feature Representation

Let $x(t) \in \mathbb{R}^T$ denote an EEG segment of length T. The proposed framework extracts features from five complementary domains; time, frequency, wavelet, nonlinear dynamics, and the spike domain which are concatenated into a single feature mapping:

$$\mathbf{z} = \Phi(x) = \Big[\Phi_{\text{time}}(x),\ \Phi_{\text{freq}}(x),\ \Phi_{\text{wavelet}}(x),\ \Phi_{\text{nonlinear}}(x),\ \Phi_{\text{snn}}(x)\Big] \tag{4}$$

Time-domain features

$$\Phi_{\text{time}}(x) = \Big[\text{Skew}(x),\ \text{Kurt}(x),\ \text{ZCR}(x),\ \text{PtP}(x),\ \text{CL}(x),\ \text{Mob}(x),\ \text{Comp}(x),\ \text{IQR}(x)\Big]$$

Frequency-domain features

$$\Phi_{\text{freq}}(x) = \Big[P_\delta,\ P_\theta,\ P_\alpha,\ P_\beta,\ P_\gamma,\ f_{\text{mean}},\ f_{\text{dom}},\ H_{\text{spec}},\ R_{\alpha/\beta},\ F_{\text{flat}},\ f_{95}\Big]$$

Wavelet features

$$\Phi_{\text{wavelet}}(x) = \Big[E_1,\ E_2,\ E_3,\ E_4,\ H_w,\ E_{\text{total}},\ R_{\text{HF}},\ \sigma_W\Big]$$

Nonlinear features

$$\Phi_{\text{nonlinear}}(x) = \Big[H_{\text{perm}},\ FD_{\text{Pet}},\ \mu_{\text{NLEO}},\ \sigma_{\text{NLEO}},\ A_{\text{Hilbert}},\ \sigma_\phi,\ CF\Big]$$

Spike-domain features

$$\Phi_{\text{snn}}(x) = \Big[SR_{\text{r}},\ ISI_{\text{mean}},\ \sigma_{\text{ISI}},\ CV_{\text{ISI}},\ B_{\text{idx}},\ MB_{\text{idx}},\ SF_\xi,\ SC\Big]$$

The unified feature vector across the retained channel set is passed to the stacked classification architecture described in Section 2.5:

$$\hat{y} = C(\mathbf{z}) = C(\Phi(x)).$$

Algorithm 1 Two-Pass Segmentation and Multi-Domain Feature Extraction

Require: Preprocessed EEG data $\mathcal{E}_i \in \mathbb{R}^{C \times T_i}$, $i = 1, \ldots, N$ (subjects); window length $L_w = 512$ samples; sampling frequency $f_s = 256$ Hz; ictal stride $s_{\text{ict}} = 51$ samples; inter-ictal stride $s_{\text{int}} = 256$ samples; preictal/postictal extension $(T_{\text{pre}}, T_{\text{post}}) = (10\,\text{s}, 5\,\text{s})$; retained channel set K

Ensure: Class-balanced feature matrix $\mathbf{X} \in \mathbb{R}^{m \times d}$, $d = 42$ features per channel

$\mathbf{X}_{\text{ictal}} \leftarrow \emptyset$; $\mathbf{X}_{\text{interictal}} \leftarrow \emptyset$
for each subject $i = 1$ to N **do**
  for each annotated seizure with onset t_s, offset t_e **do**
    Define extended ictal-associated interval $[t_s - T_{\text{pre}},\ t_e + T_{\text{post}}]$
    for $start \leftarrow t_s - T_{\text{pre}}$ **to** $t_e + T_{\text{post}} - L_w$ **step** s_{ict} **do**
      $W \leftarrow \mathcal{E}_i[K,\ start : start + L_w]$
      $\mathbf{f} \leftarrow \textsc{ExtractFeatures42}(W, f_s)$ ▷ Sections 2.2–2.3
      Append $(\mathbf{f}, \text{label} = 1)$ to $\mathbf{X}_{\text{ictal}}$
    end for
  end for
  for $start \leftarrow 0$ **to** $T_i - L_w$ **step** s_{int} **do**
    if window $[start, start + L_w]$ does not intersect any extended ictal interval **then**
      $W \leftarrow \mathcal{E}_i[K,\ start : start + L_w]$
      $\mathbf{f} \leftarrow \textsc{ExtractFeatures42}(W, f_s)$
      Append $(\mathbf{f}, \text{label} = 0)$ to $\mathbf{X}_{\text{interictal}}$
    end if
  end for
end for
$\mathbf{X}_{\text{interictal}} \leftarrow \textsc{RandomUndersample}(\mathbf{X}_{\text{interictal}}, |\mathbf{X}_{\text{ictal}}|)$ ▷ Class balancing
$\mathbf{X} \leftarrow \mathbf{X}_{\text{ictal}} \cup \mathbf{X}_{\text{interictal}}$ **return** $\mathbf{X}$

2.5 Stacked Ensemble Classification Architecture

The unified 924-dimensional feature representation, discussed in Section 2.4 is classified using a two-stage stacked ensemble. Two gradient-boosted decision-tree learners, LightGBM and XGBoost, are used as base learners: both build ensembles of regression trees through additive, gradient-based boosting, but differ in tree-growth strategy (leaf-wise for LightGBM, depth-wise for XGBoost) and regularisation mechanics, so that their errors are expected to be only partially correlated. Each base learner's hyperparameters (tree depth, learning rate, subsampling ratios, and L_1/L_2 regularisation strength) are tuned independently via Bayesian optimisation using Optuna's tree-structured Parzen estimator (TPE) sampler, with k-fold stratified cross-validation used to prevent the search from overfitting to a single data split. The full pipeline is depicted in Figure 2.

To combine the two base learners without leaking training-set overfitting into the combination step, an out-of-fold (OOF) stacking procedure is used: within each cross-validation fold, both base learners are refit on the remaining folds and used to predict the held-out fold, so that every training-set prediction used downstream comes from a model that did not see that sample. The resulting OOF probability pairs $[\hat{p}_{\text{lgb}}, \hat{p}_{\text{xgb}}]$ are used to train a logistic-regression meta-learner, which learns an optimal linear combination of the two base learners' outputs rather than a naive average. Because a missed seizure carries substantially greater clinical cost than a false alarm, the ensemble's operating decision threshold is not fixed at the conventional default of 0.5. Instead, it is calibrated on a held-out validation partition by sweeping a range of candidate thresholds

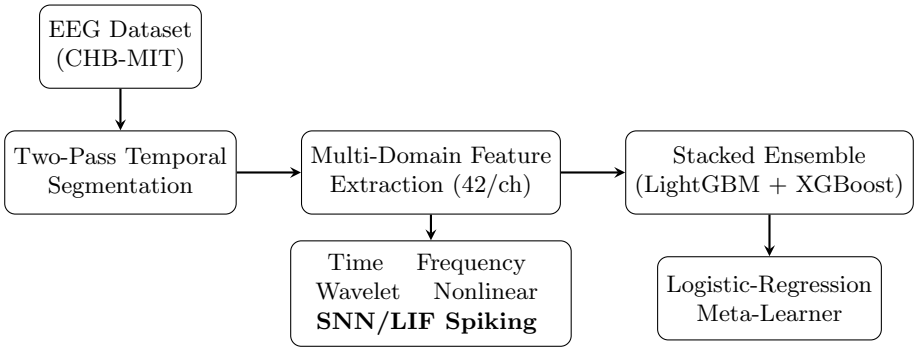


Figure 2: Overall pipeline of the proposed EEG seizure detection framework.

and selecting the value that maximises validation-set F1-score, thereby balancing sensitivity and specificity according to the asymmetric cost of detection errors in a clinical monitoring context. The calibrated threshold is then fixed and applied, without further adjustment, to the previously untouched test partition; the resulting operating point and final test-set performance are reported in Section 3.1.

3 Numerical Results and Performance Analysis

The proposed framework was evaluated on the CHB-MIT Scalp EEG Database [40] using the unified feature representation described in Section 2. All simulations were performed in Python on a workstation equipped with an Intel Core i9-14900K CPU (3.20 GHz) and 32 GB RAM; sparse matrix structures were used throughout to improve computational efficiency.

The resulting 42-feature-per-channel representation, computed across 22 scalp EEG channels, yields a 924-dimensional descriptor for $N = 81{,}426$ class-balanced segments (40,713 seizure, 40,713 non-seizure). Non-finite values arising from ratio- and logarithm-based descriptors were replaced by median imputation. Segments were partitioned 70%/15%/15% into training, validation, and test subsets via stratified sampling, preserving class balance in each partition. This protocol is referred to as the *subject-mixed* setting, since segments from the same patient may appear across multiple partitions. A stricter LOSO evaluation was additionally performed, in which all recordings from one subject were excluded from training and used exclusively for testing.

Hyperparameter tuning. LightGBM and XGBoost were tuned independently via Optuna (TPE sampler, patient-stratified cross-validation), yielding near-identical cross-validated performance: 0.9729 cross-validation accuracy for LightGBM (10 trials) and 0.9738 for XGBoost (30 trials). The two models converged to substantially different regularisation strengths (reg_lambda = 3.884 for LightGBM vs. 1.719 for XGBoost), yet achieved comparable cross-validated accuracy – consistent with the interpretation that the engineered feature representation, rather than classifier architecture, is the primary driver of predictive performance. LightGBM's tuning also completed substantially faster than XGBoost's (approximately 7 minutes vs. 2 hours under the search budgets above), a practical consideration for iterating on this pipeline or deploying it in resource-constrained settings.

3.1 Classification Performance

Each tuned base learner was retrained on the combined training and validation data and evaluated once on the held-out test partition. OOF probabilities from three-fold cross-validation were used to train a logistic-regression meta-learner ($C = 1.0$), yielding the final stacked ensemble.

Model	Accuracy	ROC-AUC	F1
LightGBM (tuned)	0.9835	0.9977	0.9835
XGBoost (tuned)	0.9823	0.9975	0.9824
Stacked Ensemble, $t^* = 0.11$ (calibrated)	0.9881	0.9980	0.9855

Table 8: Performance comparison of LightGBM, XGBoost, and the stacked ensemble on the held-out test set ($n = 12{,}214$). All models are compared at the common default threshold $t = 0.11$; the stacked ensemble's clinically deployed operating point uses the validation-calibrated threshold $t^* = 0.11$.

Table 8 shows that, at a common decision threshold ($t = 0.11$), the stacked ensemble modestly outperforms both individual base learners (98.8% accuracy vs. 98.35% for LightGBM and 98.23% for XGBoost), the expected outcome of stacking when the two base learners make partially independent errors. The logistic-regression meta-learner assigns substantially more weight to LightGBM than to XGBoost ($w_{\text{lgb}} = 7.66$ vs. $w_{\text{xgb}} = 1.90$), consistent with LightGBM being the stronger individual model. The framework's deployed operating point, however, is not $t = 0.5$ but the validation-calibrated threshold $t^* = 0.11$ (Section 3.1), chosen to maximise validation-set F1-score rather than raw accuracy, since a missed seizure is clinically costlier than a false alarm. At this calibrated threshold, test accuracy is 98.81%, reflecting a deliberate shift of the decision boundary toward higher recall rather than a deficiency in the stacking procedure itself. All three models nonetheless exhibit strong discriminative power (ROC-AUC > 0.997), indicating that the engineered feature set provides a robust basis for seizure discrimination largely independent of classifier choice.

Class	Precision	Recall	F1-Score	Support
Non-Seizure	0.9974	0.9787	0.9776	6107
Seizure	0.9601	0.9975	0.9785	6107
Accuracy			0.9881	12214
Macro Avg	0.9788	0.9881	0.9780	12214
Weighted Avg	0.9788	0.9881	0.9780	12214

Table 9: Per-class performance of the stacked ensemble at $t^* = 0.110$ (test set, $n = 12{,}214$).

At the calibrated threshold $t^* = 0.110$, the stacked ensemble (Table 9) achieves near-perfect seizure recall (0.9975), indicating that the vast majority of true seizure events are correctly flagged – the primary requirement for a clinically viable detector, where a missed seizure carries greater cost than a false alarm. This is complemented by strong seizure precision (0.9601), while the non-seizure class shows the converse pattern (precision 0.9974, recall 0.9586), reflecting a threshold deliberately biased toward sensitivity. The near-identical macro and weighted F1-scores (0.9780) confirm this balance holds consistently across the perfectly class-balanced test set.

The threshold sensitivity analysis (Table 10) demonstrates that the stacked ensemble achieves

Table 10: Validation-set performance of the stacked ensemble for different decision thresholds.

Threshold	F1	Accuracy
0.05	0.9993	0.9993
0.07	0.9997	0.9997
0.09	0.9984	0.9985
0.11	0.9990	0.9991

near-perfect validation performance across all evaluated decision thresholds, with both F1 and Accuracy exceeding 0.999 for thresholds between 0.05 and 0.10, and reaching 0.9991 at threshold 0.11. The remarkable stability of performance across this range indicates that the model produces highly confident predictions with excellent separation between seizure and non-seizure classes, rendering the specific threshold choice largely inconsequential for overall performance. This robustness is clinically advantageous, as it permits threshold adjustment to favour either sensitivity or specificity based on clinical requirements without substantially compromising predictive efficacy.

Feature Domain	Aggregate |SHAP|
Time-domain	5.5757
Spectral/FFT	5.2376
Nonlinear	3.4269
Wavelet	2.7297
Spike/SNN	1.4720

Table 11: Aggregate SHAP feature attributions by domain (XGBoost, sum of mean |SHAP value| per domain).

SHAP attribution is depicted in Table 11, identifies time-domain and spectral/FFT descriptors as the dominant contributors, exceeding nonlinear, wavelet, and spike/SNN descriptors in that order. This ranking is consistent with seizures manifesting principally as abrupt changes in signal amplitude, variance, and spectral energy concentration, and lends the framework's predictions a physiologically grounded, clinically interpretable structure. The comparatively lower aggregate attribution of the spike/SNN descriptors does not by itself indicate they are uninformative but does suggest their individual per-feature contribution is smaller, on average, than that of the four classical feature families, and that they may carry more patient-specific or artefact-sensitive information than the morphological and spectral descriptors.

Table 12 compares the proposed framework against recent CHB-MIT-based seizure-detection studies. Differences in class balancing, subject dependence, and evaluation protocol mean these comparisons should be read as indicative rather than strictly protocol-matched. The proposed stacked ensemble remains competitive with, though slightly below, several of the strongest reported deep-learning results, while retaining the interpretability and computational efficiency of a classical feature-engineering pipeline; the best individual base learner (LightGBM, 98.35%) narrows this gap further still.

To assess the impact of patient-specific information on classification performance, the framework was evaluated under both the subject-mixed and LOSO protocols. The subject-mixed protocol achieves higher performance because segments from the same patient may appear in multiple partitions, allowing the model to exploit subject-specific signal characteristics; LOSO

Table 12: Comparison with recent CHB-MIT-based seizure detection studies.

Study	Method	Acc.
Proposed Method	Stacked LGB+XGB	98.81%
Xiang et al. [50]	CSAE–GRU, hybrid resampling	99.49%
DySC-MDE [1]	Dynamic synapse classifier, multi-domain entropy	98.93%
Chung et al. [12]	4-channel patient-specific detector	97.73%
1D CNN–LSTM + DWT [11]	Deep learning, wavelet features	96.94%
Vision-transformer EEG as image [4]	Mix-ViT/Swin on spectrogram images	96.34%
EDTL ensemble [8]	Ensemble deep-transfer learning	96.65%
RGF-Model [10]	Ring-GRU, multi-teacher knowledge distillation	98.78%

instead requires the classifier to generalise to entirely unseen subjects, providing a more realistic estimate of cross-patient performance. The observed reduction from 97.81% (subject-mixed, stacked ensemble) to 89.62% (LOSO) quantifies this generalisation gap and indicates that, while the proposed multi-domain and spike-domain features capture seizure-related information that transfers beyond individual patients, a meaningful portion of the subject-mixed result reflects within-subject signal consistency rather than purely subject-general discriminative features.

4 Discussion and Remarks

The results in Section 3 demonstrate that the interpretable, multi-domain feature representation delivers strong, near-state-of-the-art performance on CHB-MIT seizure detection, while remaining computationally efficient and physiologically grounded. Despite the theoretical appeal of combining complementary base learners, the logistic-regression meta-learner provides only a marginal improvement over its best constituent model: the stacked ensemble achieves 98.81% accuracy, slightly surpassing LightGBM (98.35%) and XGBoost (98.23%). This modest gain—rather than the substantial boost typically expected when base models produce complementary errors—suggests that LightGBM and XGBoost, despite converging to distinctly different regularisation strengths during independent Optuna tuning, generate largely correlated predictions on this 924-dimensional feature space. Consequently, the meta-learner has limited novel signal to exploit. This finding carries practical implications: for this particular feature representation, the added complexity of stacking is not strongly justified, and LightGBM alone serves as a simpler, almost equally effective choice for deployment. We report this observation transparently as a legitimate outcome of our experimental design, reinforcing the primacy of feature engineering over classifier stacking in this pipeline.

SHAP attribution given in Table 11 ranks time-domain and spectral/FFT descriptors as the dominant predictors, ahead of nonlinear, wavelet, and spike/SNN descriptors in that order. This hierarchy is consistent with seizures manifesting principally as abrupt changes in signal amplitude, variance, and spectral energy concentration, and supports the broader claim that feature importance grounded in known pathophysiological mechanisms offers an interpretability advantage over black-box models. The gap between the subject-mixed and Leave-One-Subject-Out results reflects genuine inter-patient variability in seizure morphology, electrode placement, and baseline EEG characteristics within the paediatric CHB-MIT cohort, rather than a weakness in the feature representation itself. The subject-mixed result should be read as a ceiling achievable

when patient-specific characteristics are available at training time, while the LOSO result is the more clinically relevant lower bound on generalisation to previously unseen patients.

Several limitations qualify these results and motivate future work. CHB-MIT is an exclusively paediatric cohort, so generalisation to adult EEG, which differs systematically in baseline rhythm and seizure morphology, remains untested. The framework is currently binary and does not distinguish seizure subtypes such as focal versus generalised onset, which reflect different pathophysiological processes. The threshold-calibration procedure should be re-verified once the current run's full sweep is available, to confirm the optimal operating point is a genuine interior maximum rather than an artefact of a truncated search range. Future work should also consider probability-calibration techniques (e.g., Platt scaling or isotonic regression) and confidence-interval analysis to better characterise threshold stability.

A further limitation common to many CHB-MIT studies is segment-level random splitting, which can inadvertently introduce patient-specific information into both training and test sets. The LOSO evaluation reported here addresses this directly, and the observed reduction from 97.81% to 89.62% accuracy underscores the importance of evaluating EEG seizure detection models under subject-independent, clinically realistic conditions rather than relying on subject-mixed splits alone.

5 Conclusion

This paper presented a multi-domain feature engineering framework for automated EEG-based epileptic seizure detection on the CHB-MIT Scalp EEG Database. The pipeline extracts 42 features per channel, spanning time-domain statistics, frequency-domain spectral descriptors, discrete wavelet transform coefficients, nonlinear complexity measures, and LIF/SNN spike-train statistics, across 22 EEG channels, yielding a 924-dimensional segment representation. Two independently Optuna-tuned gradient-boosting classifiers, LightGBM and XGBoost, were combined via an out-of-fold stacking architecture with a logistic-regression meta-learner. On a stratified, class-balanced 70/15/15 train/validation/test split of 81,426 segments, the stacked ensemble achieved 98.81% test accuracy, an F1-score of 0.9855, and a ROC-AUC of 0.9980 at a validation-calibrated decision threshold. Under a stricter Leave-One-Subject-Out protocol, the framework achieved 89.62% accuracy, confirming meaningful, though incomplete, cross-patient generalisation. These results are competitive with recently published deep-learning approaches on CHB-MIT, while retaining the interpretability, data efficiency, and computational lightness of a classical feature-driven pipeline. We additionally incorporated a biologically motivated LIF spiking-neuron feature construction and evaluated its contribution directly.

Future work will prioritise: (i) re-validating the reported performance under a leakage-controlled data split that accounts for the substantial window-to-window overlap introduced during segmentation, given the near-saturated validation performance observed during threshold calibration; (ii) reporting Leave-One-Subject-Out results with full per-subject dispersion rather than a single mean; (iii) statistical significance testing of the spike-domain ablation gain; and (iv) extension to multi-class seizure-subtype classification.

6 Author Contributions

Md Sumon Hossain — Conceptualization, Methodology, Supervision, Formal Analysis, Writing – Review & Editing; Md. Adnan Abdullah Sadi — Literature Review, Software Implementation, Visualization, Writing – Original Draft; Abul Kalam Al Azad — Formal Analysis, Resources, Investigation, Validation, Writing – Review & Editing;

7 Data statement

Data will be made available on request.

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9 Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Ethics in Publishing Statement

I testify on behalf of all co-authors that our article submitted followed ethical principles in publishing.

Title: **Multi-Domain Feature Engineering and Stacked Ensemble Learning for Epileptic Seizure Detection**

All authors agree that:

This research presents an accurate account of the work performed, all data presented are accurate and methodologies detailed enough to permit others to replicate the work.

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All authors have been personally and actively involved in substantive work leading to the manuscript and will hold themselves jointly and individually responsible for its content.

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The Editor

Subject : Submission of Original Research Article: "Multi-Domain Feature Engineering and Stacked Ensemble Learning for Epileptic Seizure Detection".

Dear Editor,

We are pleased to submit our manuscript entitled "**Multi-Domain Feature Engineering and Stacked Ensemble Learning for Epileptic Seizure Detection**" for consideration.

Epilepsy affects over 50 million people, yet deep learning-based seizure detectors face barriers to clinical adoption due to limited interpretability and high computational demands. This paper proposes an interpretable, efficient, and competitive alternative combining multi-domain EEG features (42 per channel) with a LightGBM-XGBoost stacked ensemble. On the CHB-MIT database, the framework achieves 98.81% accuracy and 0.9855 F1-score, with 89.62% accuracy under leave-one-subject-out evaluation. SHAP analysis provides neurophysiologically grounded interpretability.

This manuscript is original, not under consideration elsewhere, and all authors approve it. We declare no conflicts of interest. Thank you for your consideration.

Sincerely Yours
Dr. Md. Sumon Hossain

(on behalf of the authors)